2.4.1 Main features of the quantum mechanical model of atom

1. The energy of electrons in atoms is quantised
2. The existence of quantized electronic energy levels is a direct result of the wave like properties of electrons. The solutions of Schrödinger wave equation gives the allowed energy levels (orbits).
3. According to Heisenberg uncertainty principle, the exact position and momentum of an electron can not be determined with absolute accuracy. As a consequence, quantum mechanics introduced the concept of orbital. Orbital is a three dimensional space in which the probability of finding the electron is maximum.
4. The solution of Schrödinger wave equation for the allowed energies of an atom gives the wave function ψ, which represents an atomic orbital. The wave nature of electron present in an orbital can be well defined by the wave function ψ.
5. The wave function ψ itself has no physical meaning. However, the probability of finding the electron in a small volume dxdydz around a point (x,y,z) is proportional to $|\psi(x,y,z)|^2$ dxdydz $|\psi(x,y,z)|^2$ is known as probability density and is always positive.

2.5 Quantum numbers

The electron in an atom can be characterised by a set of four quantum numbers, namely principal quantum number (n), azimuthal quantum number (l), magnetic quantum number (m) and spin quantum number (s). When Schrödinger equation is solved for a wave function Ψ, the solution contains the first three quantum numbers n, l and m. The fourth quantum number arises due to the spinning of the electron about its own axis. However, classical pictures of species spinning around themselves are incorrect.

Principal quantum number (n):

This quantum number represents the energy level in which electron revolves around the nucleus and is denoted by the symbol 'n'.

1. The 'n' can have the values 1, 2, 3,… n=1 represents K shell; n=2 represents L shell and n = 3, 4, 5 represent the M, N, O shells, respectively.
2. The maximum number of electrons that can be accommodated in a given shell is $2n^2$.
3. 'n' gives the energy of the electron, $E_n = \frac{(-1312.8)\,Z^2}{n^2}\ \text{kJ mol}^{-1}$ and the distance of the electron from the nucleus is given by $r_n = \frac{(0.529)n^2}{Z}\ \text{Å}$

Azimuthal Quantum number (*l*) or subsidiary quantum number :

1. It is represented by the letter '*l*', and can take integral values from zero to n-1, where n is the principal quantum number
2. Each *l* value represents a subshell (orbital). *l* = 0, 1, 2, 3 and 4 represents the s, p, d, f and g orbitals respectively.
3. The maximum number of electrons that can be accommodated in a given subshell (orbital) is $2(2l+1)$.

11th Std Chemistry 038-067.indd 44 23-12-2021 15:04:58